Depression is a major mental health disorder. Early detection is important. Traditional diagnostic methods can be subjective and time-consuming. This study used eye-tracking technology with machine learning for depression detection. Eye-tracking captures eye movements when viewing emotional images. People with depression may have longer fixations on negative stimuli. They may also show reduced attention to positive stimuli, slower saccadic movements, and pupil dilation changes. The study involved 139 young participants. They viewed positive and negative images while eye movements were recorded. Data included fixation duration, saccade velocity, blink rate, and pupil size changes. The dataset was processed and divided.